API SPECIFICATION & INTEGRATION GUIDE

IPO Allotment Checker API

Unified Multi-Registrar API for Real-Time IPO Discovery, Multi-Criteria Lookup (PAN, Application No, DP ID/Client ID), JSON Bulk & Excel Processing with Secure JWT Authentication.

Parameter	Specification Details
Production Base URL	https://api.ipoutility.in/api/v1
Root Service URL	https://api.ipoutility.in
API Version	v1.0.0 (OAS 3.1.0)
Authentication Scheme	HTTP Bearer Token (JWT / Redis Session)
Token Inactivity Timeout	30 Minutes (Rolling Inactivity Expiration)
Interactive Swagger Docs	https://api.ipoutility.in/docs
OpenAPI JSON Definition	https://api.ipoutility.in/openapi.json
Supported Registrars	Link Intime (linkintime), KFintech (kfintech), Bigshare (bigshare), Maashitla (maashitla), Skyline (skyline), Purva (purva), Cambridge (cambridge), Mas Services (masservices), Mudra (mudra), Integrated (integrated)
Data Formats	JSON (application/json), Excel (multipart/form-data)

1. Overview & Core Features

The IPO Allotment Checker API provides high-performance, normalized programmatic access to verify IPO allotment statuses across all major Indian share transfer registrars. It abstracts away registrar-specific captchas, variable HTML structures, and custom encryption routines.

- **Secure Token Authentication:** JWT Bearer authentication with single active session tracking and rolling 30-minute inactivity expiration.

- **Automated Multi-Registrar Integration:** Seamlessly connects to Bigshare, KFintech, Link Intime, Maashitla, Skyline, Purva, Cambridge, Mas Services, Mudra, and Integrated with automated payload handling and real-time response normalization.
- **Real-Time IPO Synchronization:** Dynamic discovery endpoints fetch current live IPOs declared on registrar portals without hardcoded lists.
- **Multi-Criteria Allotment Checking:** Supports checking IPO allotment status using PAN (pan), Application Number (application_no), or DP ID / Client ID / Demat (dpid_clientid).
- **Single & JSON Bulk Lookup (Max 100 Values/Request):** Perform instant single lookup via /api/v1/allotment/check-single or submit a JSON payload with up to 100 lookup values via /api/v1/allotment/check.
- **Concurrent Bulk Excel Processing:** Enables uploading Excel files (.xlsx, .xlsm) via /api/v1/allotment/check-excel for any search type (processing up to 100 unique lookup values per file); automatically parses columns, verifies them concurrently, and streams back an enriched result Excel file.

2. Authentication & Session Architecture

All protected API endpoints require authentication using a JSON Web Token (JWT) passed via the standard HTTP Bearer authorization header. Active sessions and 30-minute rolling inactivity timeouts are managed dynamically via a token session store (in-memory in single-instance deployments, and backed by Redis in multi-worker clustered deployments).

2.1 Authentication Process & Login Lifecycle

1. **Step 1 — Login Request:** The client sends a POST request to /api/v1/login with valid username and password credentials in JSON format.
2. **Step 2 — Token Issuance & Session Registration:** Upon successful credential verification, the server generates a JWT access token signed using HS256. It registers an active session in the session store (key: user_session:{username}) with a 30-minute Time-To-Live (TTL).
3. **Step 3 — Single Active Session Enforcement:** If a user logs in again while a previous session is still active, the server overwrites the Redis session key with the new token. Any existing token for that user is instantly invalidated across all devices/sessions.
4. **Step 4 — Protected Request Authorization:** The client includes the acquired token in the Authorization header of subsequent API requests: Authorization: Bearer <access_token>.
5. **Step 5 — Rolling Inactivity Timeout (30 Minutes):** Every active request to a protected endpoint validates the token against Redis. On each successful request, the system automatically resets the Redis key TTL back to 30 minutes (redis_client.expire). As long as requests continue within 30 minutes, the session stays active.
6. **Step 6 — Expiration after 30 Minutes of Inactivity:** If no API request is received for 30 consecutive minutes, the Redis session key automatically expires and is evicted. Any further

API calls with that token will fail with HTTP 401 Unauthorized: 'Session expired due to 30 minutes of inactivity. Please log in again.'

7. **Step 7 — Token Renewal:** When a token expires due to 30 minutes of no use (or is invalidated by a new login), the client application must initiate a new login call (POST /api/v1/login) to generate a fresh token.

> **[30-MINUTE INACTIVITY RULE & RE-AUTHENTICATION]** Tokens automatically expire after 30 minutes of inactivity (no API calls). Upon receiving HTTP 401 Unauthorized ('Session expired due to 30 minutes of inactivity. Please log in again.'), your client application should automatically trigger a new POST /api/v1/login call to create a new access token.

3. API Integration Workflow

Integrating the API within your client portal, mobile app, or internal ERP follows a 4-step lifecycle:

8. **Step 1 — Authenticate & Acquire Token:** Invoke POST /api/v1/login with username and password to get your Bearer access_token.
9. **Step 2 — Discover Registrars:** Invoke GET /api/v1/registrars (with Bearer token) to obtain active registrar codes (e.g. bigshare, kfintech, linkintime, maashitla, skyline, purva, cambridge, masservices, mudra, integrated).
10. **Step 3 — Fetch Live IPOs:** Invoke GET /api/v1/registrars/{registrar_id}/ipos to populate dropdowns with live IPO IDs and names.
11. **Step 4A — Single Record Allotment Check:** Submit POST /api/v1/allotment/check-single with registrar, ipo_id, ipo_name, search_type ('pan', 'application_no', or 'dpid_clientid'), and search_value in the JSON payload.
12. **Step 4B — Check JSON Bulk Array (Max 100 Values):** Submit POST /api/v1/allotment/check with registrar, ipo_id, ipo_name, search_type, and values (array of up to 100 lookup strings) in the JSON payload.
13. **Step 4C — Check Bulk Excel File:** Submit POST /api/v1/allotment/check-excel with multipart/form-data containing the Excel file, metadata, and search_type. The API streams back the formatted result workbook.

4. API Endpoints Specification

4.1 Service Health & Information (Public)

Public endpoints to verify API availability, uptime, and documentation.

Method	Endpoint URL	Description & Response
GET	https://api.ipoutility.in/	Root welcome message & doc links. Returns: 200 OK (JSON)
GET	https://api.ipoutility.in/health	Liveness probe for monitoring. Returns: {"status": "ok"}

4.2 User Authentication & Token Generation (Public)

POST https://api.ipoutility.in/api/v1/login

Authenticates user credentials, revokes any prior active session, registers a 30-minute session in Redis, and returns a new Bearer JWT access token.

Body Parameter	Type	Required	Description
username	string	Yes	Registered API username.
password	string	Yes	User authentication password.

Sample Login Request Payload (JSON):

```
{
  "username": "admin",
  "password": "admin@1234"
}
```

Sample Successful Login Response (200 OK):

```
{
  "success": true,
  "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
  "token_type": "bearer",
  "expires_in_minutes": 30
}
```

Sample Invalid Credentials Error (401 Unauthorized):

```
{
  "detail": "Invalid credentials"
}
```

4.3 List Supported Registrars (Protected)

GET https://api.ipoutility.in/api/v1/registrars

Header Required: Authorization: Bearer <access_token>

Returns all currently enabled registrars in the engine.

```
// Example Response: 200 OK
{
  "success": true,
  "count": 10,
  "data": [
    { "id": "bigshare", "name": "Bigshare" },
    { "id": "kfintech", "name": "Kfintech" },
    { "id": "linkintime", "name": "Link Intime" },
    { "id": "maashitla", "name": "Maashitla" },
    { "id": "skyline", "name": "Skyline" },
    { "id": "purva", "name": "Purva" },
    { "id": "cambridge", "name": "Cambridge" },
    { "id": "masservices", "name": "Mas Services" },
    { "id": "mudra", "name": "Mudra" },
    { "id": "integrated", "name": "Integrated" }
  ]
}
```

4.4 Get Active IPOs for a Registrar (Protected)

GET https://api.ipoutility.in/api/v1/registrars/{registrar_id}/ipos

Header Required: Authorization: Bearer <access_token>

Parameter	Type	Location	Description

registrar_id	string	Path	Registrar code: 'bigshare', 'kfintech', 'linkintime', 'maashitla', 'skyline', 'purva', 'cambridge', 'masservices', 'mudra', or 'integrated' (Required).

```
// Example Response: 200 OK
{
  "success": true,
  "registrar": "linkintime",
  "count": 2,
  "data": [
    {
      "ipo_id": "1234",
      "ipo_name": "ABC Tech Limited - IPO"
    },
    {
      "ipo_id": "1235",
      "ipo_name": "XYZ Logistics Limited - SME IPO"
    }
  ]
}
```

4.5 Single Record Allotment Check (Protected)

POST https://api.ipoutility.in/api/v1/allotment/check-single

Header Required: Authorization: Bearer <access_token>

Validates and checks allotment status for a single record in real time by PAN, Application Number, or DP ID / Client ID.

Field	Type	Required	Description
registrar	string	Yes	Target registrar ID ('bigshare', 'kfintech', 'linkintime', 'maashitla', 'skyline', 'purva',

			'cambridge', 'masservices', 'mudra', 'integrated').
ipo_id	string	Yes	The exact ipo_id returned by the active IPOs endpoint.
ipo_name	string	Yes	The exact ipo_name returned by the active IPOs endpoint.
search_type	string	Yes	Lookup criteria: 'pan', 'application_no', or 'dpid_clientid'.
search_value	string	Yes	The input lookup value (PAN / Application No / DP ID + Client ID).

> **[KFINTECH APPLICATION NUMBER RESTRICTION & TECHNICAL RATIONALE]** KFintech does not support allotment checking by Application Number (search_type: application_no). Submitting search_type: application_no for KFintech returns an HTTP 400 Bad Request error ('KFintech does not support allotment checking by Application Number. Please use PAN or DP ID / Client ID.').
>
> Technical Rationale: Upstream KFintech requires submitting both Application Number AND PAN together as mandatory dual inputs. Since PAN is already mandatory for that upstream query, requiring 2 input fields is redundant when a user can simply query directly using search_type: pan with a single input field. Therefore, for KFintech, use search_type: pan or search_type: dpid_clientid.

Sample Request Payload — Check by PAN (JSON):

```
{
  "registrar": "linkintime",
  "ipo_id": "1234",
  "ipo_name": "ABC Tech Limited - IPO",
  "search_type": "pan",
  "search_value": "ABCDE1234F"
}
```

Sample Request Payload — Check by Application Number (JSON):

```
{
  "registrar": "linkintime",
  "ipo_id": "1234",
  "ipo_name": "ABC Tech Limited - IPO",
  "search_type": "application_no",
  "search_value": "10098234"
}
```

Sample Request Payload — Check by DP ID / Client ID (JSON):

```
{
  "registrar": "kfintech",
  "ipo_id": "5678",
  "ipo_name": "XYZ Tech Limited - IPO",
  "search_type": "dpid_clientid",
  "search_value": "1208160012345678"
}
```

Sample Response — Single Application (Shares Allotted):

```
{
  "success": true,
  "status": "allotted",
  "data": [
    {
      "search_type": "pan",
      "search_value": "ABCDE1234F",
      "pan": "ABCDE1234F",
      "registrar": "linkintime",
      "ipo_id": "1234",
      "ipo_name": "ABC Tech Limited - IPO",
      "status": "allotted",
      "applicant_name": "JOHN DOE",
      "application_number": "10098234",
      "dp_id_client_id": "1208160012345678",
      "shares_applied": 150,
      "shares_allotted": 150,
      "category": "IND",
```

```
      "error_reason": ""
    }
  ]
}
```

Sample Response (Not Allotted):

```
{
  "success": true,
  "status": "not_allotted",
  "data": [
    {
      "search_type": "pan",
      "search_value": "ABCDE1234F",
      "pan": "ABCDE1234F",
      "registrar": "linkintime",
      "ipo_id": "1234",
      "ipo_name": "ABC Tech Limited - IPO",
      "status": "not_allotted",
      "applicant_name": "JOHN DOE",
      "application_number": "10098234",
      "dp_id_client_id": "1208160012345678",
      "shares_applied": 150,
      "shares_allotted": 0,
      "category": "IND",
      "error_reason": ""
    }
  ]
}
```

[MULTI-APPLICATION DYNAMIC GROUPING (SINGLE CHECK)] When an applicant has multiple bids or applications (for example, applying across multiple categories such as Retail and NII/HNI, or submitting multiple applications under one PAN as commonly found on Bigshare), the API dynamically groups the records under the SingleCheckData model. Common lookup parameters (search_type, search_value, pan, registrar, ipo_id, ipo_name) are hoisted to the top level, and all matching bid records are listed inside an 'applications' array.

Sample Response — Multi-Application Grouped Record (2+ Bids Found):

```
{
  "success": true,
  "status": "allotted",
  "data": {
    "search_type": "pan",
    "search_value": "ABCDE1234F",
    "pan": "ABCDE1234F",
    "registrar": "bigshare",
    "ipo_id": "1234",
    "ipo_name": "ABC Tech Limited - IPO",
    "applications": [
      {
        "status": "allotted",
        "applicant_name": "JOHN DOE",
        "application_number": "10098234",
        "dp_id_client_id": "1208160012345678",
        "shares_applied": 150,
        "shares_allotted": 150,
        "category": "IND",
        "error_reason": ""
      },
      {
        "status": "not_allotted",
        "applicant_name": "JOHN DOE",
        "application_number": "10098235",
        "dp_id_client_id": "1208160012345678",
        "shares_applied": 300,
        "shares_allotted": 0,
        "category": "NII",
        "error_reason": ""
      }
    ]
  }
}
```


Status Value	Explanation
allotted	The application was found and shares were successfully allotted (shares_allotted > 0).
not_allotted	The application was found on the registrar, but 0 shares were allotted.

not_found	No application or bidding record exists on the registrar for the given lookup value.
error	Registrar lookup failed due to network timeout, captcha challenge, or upstream issue.

4.6 JSON Bulk Allotment Check (Protected)

POST https://api.ipoutility.in/api/v1/allotment/check

Header Required: Authorization: Bearer <access_token>

Checks allotment status for an array of lookup values (PANs, Application Numbers, or DP ID/Client IDs) passed in a single JSON request body. Enforces a strict limit of maximum 100 lookup values per request payload.

Field	Type	Required	Description
registrar	string	Yes	Target registrar ID ('bigshare', 'kfintech', 'linkintime', 'maashitla', 'skyline', 'purva', 'cambridge', 'masservices', 'mudra', 'integrated').
ipo_id	string	Yes	The exact ipo_id returned by the active IPOs endpoint.
ipo_name	string	Yes	The exact ipo_name returned by the active IPOs endpoint.
search_type	string	Yes	Lookup criteria: 'pan', 'application_no', or 'dpid_clientid'.
values	array[string]	Yes	List of lookup values. Min length: 1, Max length: 100 items per request.

> **[100 LOOKUP VALUE SINGLE REQUEST LIMIT]** A maximum of 100 lookup items is allowed in the 'values' array per request (min_length=1, max_length=100). Submitting a JSON payload with more than 100 items will trigger an HTTP 422 Unprocessable Entity validation error. For larger lists, split items into client-side batches of 100 per request or use the /check-excel endpoint.

Sample Request Payload (JSON):

```
{
  "registrar": "linkintime",
  "ipo_id": "1234",
  "ipo_name": "ABC Tech Limited - IPO",
  "search_type": "pan",
  "values": [
    "ABCDE1234F",
    "FGHIJ5678K",
    "KLMNO9012P"
  ]
}
```

Sample Response Payload (200 OK):

```
{
  "success": true,
  "total": 3,
  "processed": 3,
  "summary": {
    "allotted": 1,
    "not_allotted": 1,
    "not_found": 1,
    "error": 0
  },
  "results": [
    {
      "search_type": "pan",
      "search_value": "ABCDE1234F",
      "pan": "ABCDE1234F",
      "registrar": "linkintime",
      "ipo_id": "1234",
      "ipo_name": "ABC Tech Limited - IPO",
      "status": "allotted",
```

```
      "applicant_name": "JOHN DOE",
      "application_number": "10098234",
      "dp_id_client_id": "1208160012345678",
      "shares_applied": 150,
      "shares_allotted": 150,
      "category": "IND",
      "error_reason": ""
    },
    {
      "search_type": "pan",
      "search_value": "FGHIJ5678K",
      "pan": "FGHIJ5678K",
      "registrar": "linkintime",
      "ipo_id": "1234",
      "ipo_name": "ABC Tech Limited - IPO",
      "status": "not_allotted",
      "applicant_name": "JANE SMITH",
      "application_number": "10098235",
      "dp_id_client_id": "1208160087654321",
      "shares_applied": 150,
      "shares_allotted": 0,
      "category": "IND",
      "error_reason": ""
    },
    {
      "search_type": "pan",
      "search_value": "KLMNO9012P",
      "pan": "KLMNO9012P",
      "registrar": "linkintime",
      "ipo_id": "1234",
      "ipo_name": "ABC Tech Limited - IPO",
      "status": "not_found",
      "applicant_name": "",
      "application_number": "",
      "dp_id_client_id": "",
      "shares_applied": 0,
      "shares_allotted": 0,
      "category": "",
      "error_reason": "No record found for the given lookup value."
    }
  ]
}
```

> **[MULTI-BID RECORDS IN BULK RESULTS]** If any individual lookup value in the bulk list matches multiple application bids, its entry inside the 'results' array will adopt the grouped SingleCheckData format with an inner 'applications' array, while single-bid records remain as flat objects.

4.7 Bulk Excel Allotment Check (Protected)

POST https://api.ipoutility.in/api/v1/allotment/check-excel

Header Required: Authorization: Bearer <access_token>

Uploads an Excel spreadsheet of lookup values (PANs, Application Numbers, or DP ID/Client IDs), executes concurrent allotment queries across workers, and streams back a fully formatted, downloadable .xlsx file with enriched results. Processes up to 100 unique lookup values per uploaded workbook.

Content-Type: multipart/form-data
Response Type: application/vnd.openxmlformats-officedocument.spreadsheetml.sheet

Form Field	Type	Required	Description
file	File (binary)	Yes	Excel workbook (.xlsx or .xlsm) containing lookup values. Up to 100 unique values are extracted and processed.
registrar	string	Yes	Target registrar ID ('bigshare', 'kfintech', 'linkintime', 'maashitla', 'skyline', 'purva', 'cambridge', 'masservices', 'mudra', 'integrated').
ipo_id	string	Yes	The target ipo_id from active IPOs.
ipo_name	string	Yes	The target ipo_name from active IPOs.

search_type	string	Yes	Lookup criteria: 'pan', 'application_no', or 'dpid_clientid'.

> **[100 RECORDS LIMIT & STREAMING DOWNLOAD]** The Excel parser extracts and verifies up to 100 unique lookup values per file. The response is a direct binary file stream with header Content-Disposition: attachment; filename="<IPO_NAME>_allotment_result.xlsx". If validation fails (e.g. no valid lookup values found), the API returns a JSON error response.

5. Excel File Specifications

5.1 Accepted Header Names for Input

The Excel parser features fuzzy header recognition to comfortably handle varied client formats:

Lookup Search Type	Accepted Header Examples (Case-Insensitive)
PAN (search_type: pan)	PAN, Pan, pan, PAN NO, PAN_NO, PAN-NO, PAN Number, PAN Card
Application No (search_type: application_no)	Application No, App No, Application Number, App_No, AppNo, Application
DP ID / Client ID (search_type: dpid_clientid)	DP ID, Client ID, DPID_CLIENTID, Demat, Demat Account, BO ID, DP/Client ID
Headerless Spreadsheets	If no recognized header is found, the parser processes Column A from Row 1 as raw lookup values.

5.2 Output Spreadsheet Structure

The downloaded Excel result file uses a wide multi-application format. Column A dynamically reflects the lookup criteria, and columns B onwards contain structured blocks for each application record (numbered Application Num1, DP_ID1, Category1, etc.), expanding horizontally if an entity submitted multiple bids:

Column / Block	Header Label	Description & Example Value

Col A (Dynamic)	PAN / DP ID / Application No	Standardized lookup identifier searched (e.g. ABCDE1234F).
Col B (App 1)	Application Num1	Registrar application reference number (e.g. 62162676).
Col C (App 1)	DP_ID1	Demat account ID (16-digit CDSL or NSDL 8+8).
Col D (App 1)	Category1	Investor category (e.g. IND, RETAIL, NII, HNI).
Col E (App 1)	Name1	Applicant name registered on application (e.g. JOHN DOE).
Col F (App 1)	Applied1	Number of shares bid / applied for (e.g. 150).
Col G (App 1)	Alloted1	Number of shares allotted (e.g. 150 or 0).
Col H (App 1)	Application Status1	Normalized status: 'ALLOTEE', 'NON-ALLOTEE', or 'No data found'.
Col I (App 1)	Error Reason1	Detailed error message if lookup failed, otherwise blank.
Cols J+ (App 2..N)	Application Num2, DP_ID2...	Repeats the 8 application columns side-by-side if multiple bids exist.

6. Error Codes & Exception Handling

All error responses follow a standardized JSON envelope with consistent machine-readable codes:

```
{
  "success": false,
  "error": {
    "code": "ERROR_CODE_STRING",
    "message": "Human readable explanation of the failure."
  }
}
```

HTTP Status	Error Detail / Message	Description & Resolution
401 Unauthorized	Session expired due to 30 minutes of inactivity. Please log in again.	30 minutes of inactivity reached; Redis session key expired. Client must call POST /api/v1/login to create a new token.
401 Unauthorized	Session invalidated. User logged in from another device/session.	User logged in from another session/device, overwriting active Redis session. Obtain new token.
401 Unauthorized	Invalid credentials	Username or password in POST /api/v1/login is incorrect.
400 Bad Request	KFintech does not support allotment checking by Application Number...	KFintech requires mandatory PAN for application lookups. Use search_type 'pan' or 'dpid_clientid'.
400 Bad Request	REGISTRAR_NOT_SUPPORTED	The registrar_id specified does not match supported registrars.
400 Bad Request	INVALID_PAN	The PAN format is invalid. Must match [A-Z]{5}[0-9]{4}[A-Z].
400 Bad Request	NO_VALID_LOOKUP_VALUE	The uploaded Excel file does not contain any valid lookup values for the specified search_type.
403 Forbidden	LICENSE_EXPIRED	Client license or subscription is expired.
422 Unprocessable	VALIDATION_ERROR / List max length	FastAPI schema validation error. Triggers if 'values' array in single JSON bulk request exceeds 100 items limit.
500 Internal	INTERNAL_ERROR	Unhandled server exception. Check parameters and retry.

7. Operational Best Practices

- **Automated Token Refresh on 401:** Configure HTTP interceptors in client applications to listen for HTTP 401 errors. If a 401 with session expiration is received after 30 minutes of inactivity, trigger POST /api/v1/login to re-authenticate seamlessly.

- **Preferred Search Types across Registrars:** Use search_type='pan' or search_type='dpid_clientid' as primary lookup choices, since they are supported uniformly across all registrars including KFintech.
- **Single JSON Request Limit (100 Items):** When calling /api/v1/allotment/check with JSON, pass at most 100 lookup items in the 'values' array per request payload (enforced by Pydantic max_length=100). For larger datasets, split items into client-side chunks of 100 per request.
- **Concurrency Management:** For bulk checks, keep concurrency between 3 and 7 (default is 5). Excessively high concurrency can trigger temporary IP throttling on registrar web servers.
- **Batch File Sizing for Excel:** The API processes up to 100 unique lookup values per uploaded Excel spreadsheet (matching the 100-item batch ceiling). For larger datasets, batch your Excel files into increments of up to 100 records per file for optimal turnaround time.
- **Dynamic Dropdown Caching:** Cache the active IPOs list locally for 2-5 minutes to prevent redundant registrar scraping on repeated UI renders.
- **Handling Transient Errors:** If an individual record returns status='error', it typically indicates a transient connection timeout or high load on the registrar. Such records can be safely retried.

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